BC LiDAR · road-planning example

Access road · entered design assumptions

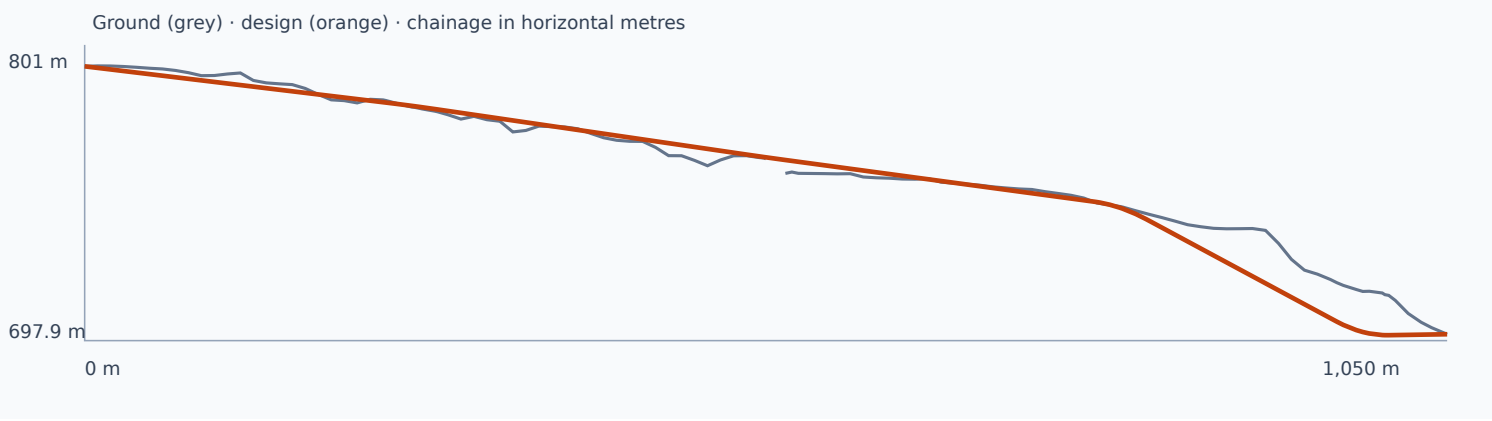
Partial model · totals cover supported intervals only
96.3% coverage (1,011.3 of 1,050 m). Source terrain supports preliminary geometry; accuracy is not independently surveyed.

Length	Cut	Fill	Scenario cost
1,050 m	**54,259.8 m³**	**7,951.4 m³**	**$4,502,091**

Terrain and assumptions

Source	BC 093G086 · classified-ground LiDAR DTM · 1 m (July 2024)
Native/display spacing	1 m; source grid resolution, not vertical accuracy
Horizontal CRS	EPSG:3157
Vertical reference	CGVD2013 (CGG2013), EPSG:6647; metres
Source URL	https://pgmaps.ahmadjalil.com/data/lidar-analysis/bc-093g086/dtm-1m.tif
SHA-256	b1259f310ceed97a11f25883d7df609390b977a293b622f9e4946a7853ad00e2
Sampling interval	10 m requested; exact event stations and computation limits can alter spacing
Review thresholds	Grade 12%; minimum radius 25 m (user assumptions)
Cost assumptions	Cut $12/m³; fill $8/m³; surfacing $40/m³; clearing $1.5/m²; reusable cut factor 0.9

Longitudinal profile



Engineering cost breakdown

Item	CAD
cut	$2,937,217.95
stripping	$0

Item	CAD
fill	$63,611.12
surfacing	$67,278.64
clearing	$22,471.6
structures	$8,541.82
haul	$115,118.03
borrow	$96,146.83
waste	$588,475.37
sitePreparation	$4,000
mobilization	$12,000
subtotal	$3,914,861.34
contingency	$587,229.2
total	$4,502,090.54

Material quantities

Ground horizons are entered assumptions or measurements. They are not inferred from the terrain model. Excavated stripping is classified cut only; undercut beneath fill is not included.

Material	Bank cut m³	Usable compacted m³	Unusable bank m³	Excavation CAD
Till · assumed	12,546.1	8,468.6	3,136.5	225,829.75
Rock · assumed	41,713.7	16,059.8	27,113.9	2,711,388.2

Culvert geometry

Name	Station m	Grade %	Minimum cover m	Required horizontal span m	Inclined barrel m	Flags
Cross-drain A · example	300	-2.27	2.03	12.1	22	

Haul scenario

Deterministic nearest-source allocation within covered runs enabled. Borrow: 3,004.6 compacted m³; waste: 49,039.6 bank m³; cut/fill and borrow haul: 2,317.7 compacted m³·km; waste haul: 13,170.4 bank m³·km. Unallocated borrow: 9.7 compacted m³; unallocated waste: 13,472 bank m³. Haul distances follow road chainage plus site access; this is not a global haul optimizer.

- Haul is allocated separately within each continuously supported road section; material is not moved across a terrain/daylight gap.
- Some borrow/waste has no reachable site capacity. Default supply/disposal rates are included, but its unknown haul distance and haul cost are excluded.

Quantity basis and limitations

Cut/fill areas are integrated from sampled cross-sections; interval volumes use average end areas. No-data or unclosed daylight sections leave an interval incomplete. Mass balance applies entered material reuse and compaction factors, or the default reusable-cut factor for unclassified ground. Costs use the entered rates, structures, haul scenario, mobilization and contingency where supplied; zero rates remain unpriced. Geotechnical treatments are not modeled. Ground is held separate from designed road elevation.

This is a preliminary planning record. It does not establish construction suitability, slope stability, drainage capacity, fish passage, permits or cable operating safety. D-lines screen assumed geometry without cable tension, load or anchor analysis. Crossing intersections require review. Entered culverts provide geometry and cover checks, not a hydraulic or fish-passage design.

Model flags

- Terrain coverage or daylight is incomplete. Quantities and cost are covered subtotals; the mass curve stops at the first gap.
- 24 stations exceed the 12% grade limit.
- Haul is allocated separately within each continuously supported road section; material is not moved across a terrain/daylight gap.
- Some borrow/waste has no reachable site capacity. Default supply/disposal rates are included, but its unknown haul distance and haul cost are excluded.

Crossing review

No intersections with the supplied context were found. This does not establish the absence of streams or constraints.

Station schedule

Station m	Ground m	Design m	Grade %	Cut m²	Fill m²	Mass m³		Review flags
0	798.86	798.86	-5.74	1.05	0	0	Complete	
10	799.03	798.29	-5.74	7.08	0	27.5	Complete	
20	798.99	797.71	-5.74	12.4	0	92.8	Complete	
30	798.8	797.14	-5.74	16.44	0	186.4	Complete	
40	798.51	796.56	-5.74	19.32	0	297	Complete	
50	798.16	795.99	-5.74	23.37	0	423.3	Complete	
60	797.9	795.42	-5.74	25.89	0	564.4	Complete	
70	797.35	794.84	-5.74	25.46	0	710	Complete	
80	796.53	794.27	-5.74	23.45	0	850.2	Complete	
90	795.42	793.69	-5.74	18.61	0	975.7	Complete	
100	795.49	793.12	-5.74	24.23	0	1,103	Complete	
110	796.05	792.55	-5.74	38.75	0	1,273.8	Complete	
120	796.43	791.97	-5.74	49.94	0	1,499.5	Complete	
130	793.67	791.4	-5.74	27.58	0	1,701.9	Complete	

Station m	Ground m	Design m	Grade %	Cut m²	Fill m²	Mass m³	Review flags
140	792.78	790.82	-5.74	20.01	0	1,839.9	Complete
150	792.43	790.25	-5.74	21.84	0	1,964.2	Complete
160	792.14	789.67	-5.74	24.05	0	2,097.2	Complete
170	790.73	789.1	-5.74	15.06	0	2,213.5	Complete
180	788.52	788.53	-5.74	3.75	3.52	2,255.9	Complete
190	786.54	787.95	-5.74	0	9.41	2,203.8	Complete
200	786.24	787.38	-5.74	0	8.98	2,111.8	Complete
210	785.41	786.8	-5.74	0	12.9	2,002.4	Complete
220	786.74	786.23	-5.74	4.64	8.23	1,912.4	Complete
230	786.52	785.66	-5.74	8.74	2.15	1,905.7	Complete
232	786.31	785.54	-5.74	8.87	0.44	1,915	Complete
240	785.29	785.07	-5.99	7.46	0	1,957.3	Complete
250	784.11	784.46	-6.3	2	1.92	1,979.6	Complete
252	783.89	784.33	-6.36	1.16	2.98	1,976.9	Complete
253.9	783.8	784.21	-6.42	1.34	2.89	1,972.8	Complete
258.4	783.38	783.92	-6.55	0.35	3.58	1,960.9	Complete
260	783.19	783.81	-6.6	0	4.4	1,954.7	Complete
270	782.35	783.14	-6.91	0	5.7	1,904.2	Complete
272	782.13	783	-6.97	0	6.77	1,891.7	Complete
280	781.04	782.44	-6.97	0	14.44	1,806.8	Complete
290	779.45	781.74	-6.97	0	22.15	1,623.9	Complete
300	780.46	781.05	-6.97	0	6.08	1,482.8	Complete
310	779.22	780.35	-6.97	0	10.74	1,398.7	Complete
320	778.66	779.65	-6.97	0	10.58	1,292	Complete
330	774.73	778.95	-6.97	0	57.36	952.3	Complete
340	775.27	778.26	-6.97	0	35.67	487.1	Complete
350	776.86	777.56	-6.97	0	5.75	280	Complete
360	776.67	776.86	-6.97	0.44	1.08	247.3	Complete
370	776.55	776.17	-6.97	6.04	0	263.8	Complete
380	775.88	775.47	-6.97	4.85	0.67	297.2	Complete
390	774.19	774.77	-6.97	1.24	4.67	291	Complete
400	772.58	774.07	-6.97	0	13.86	202.5	Complete
410	771.66	773.38	-6.97	0	17.36	46.4	Complete
420	771.28	772.68	-6.97	0	13.02	-105.6	Complete
430	771.24	771.98	-6.97	0	8.06	-210.9	Complete
440	769.02	771.29	-6.97	0	25.52	-378.8	Complete

Station m	Ground m	Design m	Grade %	Cut m²	Fill m²	Mass m³	Review flags
450	766	770.59	-6.97	0	55.38	-783.3	Complete
460	765.96	769.89	-6.97	0	45.57	-1,288	Complete
470	764.25	769.19	-6.97	0	54.01	-1,785.9	Complete
480	762.29	768.5	-6.97	0	67.59	-2,393.9	Complete
490	764.41	767.8	-6.97	0	17.48	-2,819.2	Complete
500	765.94	767.1	-6.97	0	7.88	-2,946	Complete
505	765.97	766.76	-6.97	0	4.81	-2,977.6	Complete
510	765.98	766.41	-6.91	0	2.09	-2,994.9	Complete
520	765.28	765.72	-6.77	—	—	—	Right daylight unresolved: extend section or check terrain; Missing terrain inside the road footprint
523.7	765.11	765.48	-6.72	1.81	7.21	—	Complete
525	764.86	765.39	-6.71	1.12	9.54	—	Complete
529.8	—	765.07	-6.64	—	—	—	Left daylight unresolved: extend section or check terrain; Missing terrain inside the road footprint
530	—	765.05	-6.64	—	—	—	Left daylight unresolved: extend section or check terrain; Missing terrain inside the road footprint
540	759.47	764.4	-6.51	2.66	43.05	—	Complete
545	759.97	764.07	-6.44	3.5	33.81	—	Complete
550	759.49	763.75	-6.44	0	38.46	—	Complete
560	759.44	763.11	-6.44	0	41.28	—	Complete
570	759.39	762.46	-6.44	0	32.38	—	Complete
580	759.3	761.82	-6.44	0	24.69	—	Complete
590	759.36	761.17	-6.44	0	17.6	—	Complete
600	758.15	760.53	-6.44	0	22.15	—	Complete
610	757.85	759.89	-6.44	0	17.69	—	Complete
620	757.71	759.24	-6.44	0	11.85	—	Complete
630	757.4	758.6	-6.44	0	8.87	—	Complete
640	757.33	757.95	-6.44	0	3.88	—	Complete
650	757.58	757.31	-6.44	3.29	0	—	Complete
660	756.25	756.67	-6.44	0	2.46	—	Complete
670	755.86	756.02	-6.44	0	0.72	—	Complete
680	755.48	755.38	-6.44	1.86	0	—	Complete
690	755.12	754.73	-6.44	4.22	0	—	Complete
700	754.52	754.09	-6.44	4.64	0	—	Complete
710	754.11	753.45	-6.44	6.81	0	—	Complete
720	753.75	752.8	-6.44	9.29	0	—	Complete
730	753.55	752.16	-6.44	13.63	0	—	Complete

Station m	Ground m	Design m	Grade %	Cut m²	Fill m²	Mass m³	Review flags
740	752.76	751.51	-6.44	12.23	0	—	Complete
750	752.1	750.87	-6.44	12.7	0	—	Complete
760	751.43	750.23	-6.44	12.58	0	—	Complete
770	750.4	749.58	-6.44	8.24	0	—	Complete
776.4	749.22	749.17	-6.44	1.36	0.13	—	Complete
778	749	749.07	-6.44	0.81	0.76	—	Complete
780	748.59	748.93	-7.41	0	1.94	—	Complete
782.5	748.41	748.73	-8.59	0	2.03	—	Complete
790	747.87	747.95	-12.21	0.35	0.26	—	Grade exceeds 12%
798	747.28	746.82	-16.05	4.79	0	—	Grade exceeds 12%
800	747.08	746.48	-17.02	5.88	0	—	Grade exceeds 12%
810	745.73	744.54	-21.83	12.43	0	—	Grade exceeds 12%
818	744.71	742.65	-25.66	21.07	0	—	Grade exceeds 12%
820	744.45	742.13	-25.66	24.07	0	—	Grade exceeds 12%
830	743.23	739.56	-25.66	41.29	0	—	Grade exceeds 12%
840	741.95	737	-25.66	59.82	0	—	Grade exceeds 12%
850	740.59	734.43	-25.66	82.98	0	—	Grade exceeds 12%
860	739.86	731.86	-25.66	113.76	0	—	Grade exceeds 12%
870	739.26	729.3	-25.66	157.56	0	—	Grade exceeds 12%
880	739.09	726.73	-25.66	216.52	0	—	Grade exceeds 12%
890	739.12	724.16	-25.66	290.43	0	—	Grade exceeds 12%
900	739.18	721.6	-25.66	372.99	0	—	Grade exceeds 12%
910	738.5	719.03	-25.66	432.27	0	—	Grade exceeds 12%
920	733.73	716.46	-25.66	356.97	0	—	Grade exceeds 12%
930	727.84	713.9	-25.66	277.99	0	—	Grade exceeds 12%
940	723.87	711.33	-25.66	270.12	0	—	Grade exceeds 12%
950	722.41	708.76	-25.66	269.4	0	—	Grade exceeds 12%
960	720.44	706.2	-25.66	286.97	0	—	Grade exceeds 12%
965	719.26	704.92	-25.66	290.53	0	—	Grade exceeds 12%
970	718.25	703.72	-22.35	295.62	0	—	Grade exceeds 12%
980	716.75	701.81	-15.74	301.75	0	—	Grade exceeds 12%
985	716.04	701.11	-12.45	—	—	—	Missing terrain inside the road footprint; Grade exceeds 12%
990	716.1	700.57	-9.14	294.78	0	—	Complete
990	716.11	700.57	-9.13	294.93	0	—	Complete
1,000	715.47	699.98	-2.52	352.75	0	—	Complete
1,002	714.85	699.95	-1.21	361.25	0	—	Complete

Station m	Ground m	Design m	Grade %	Cut m²	Fill m²	Mass m³	Review flags
1,005	714.59	699.94	0.77	343.21	0	—	Complete
1,010	712.76	699.98	0.77	303.14	0	—	Complete
1,020	707.92	700.06	0.77	198.47	0	—	Complete
1,030	704.68	700.13	0.77	117.57	0	—	Complete
1,038	702.72	700.19	0.77	66.7	0	—	Complete
1,040	702.3	700.21	0.77	51.84	0	—	Complete
1,050	700.29	700.29	0.77	1.31	0.02	—	Complete

Project schema: pgmaps-road-project-v1. Export the project JSON for editable geometry and the CSV for numeric station values. Geographic export coordinates are WGS84 longitude/latitude; design elevations retain the recorded terrain vertical reference.